Code Dynamos

WEB WEAVE '26

Problem Statement

A full-featured Club Hub — Events · Challenges · Team · Gallery · Projects

Build | Compete | Connect | Innovate | Deploy

1. Overview

Participants must design and develop a comprehensive **Club Hub Web Application** for their college technical club. This platform goes beyond a typical competitive programming site - it serves as the **official digital identity** of the club, combining member management, event hosting, challenge systems, project showcases, and a rich media gallery into one cohesive, modern web experience.

Inspired by platforms such as LeetCode, HackerRank, and Devfolio, the application must incorporate club-centric features that make it the go-to hub for every member from curious freshers to experienced seniors.

Club Identity	Real-time Leaderboard	Event Management
Member Profiles	Project Showcase	Media Gallery

2. Problem Statement

College technical clubs often lack a **centralised, professional web presence**. Activities are scattered across WhatsApp groups, Google Forms, and social media pages, making it difficult for students to discover events, track their growth, or understand what the club has achieved. Participants must solve the following key pain points:

No single source of truth	Club info is fragmented - members cannot find team details, past projects, or results in one place.
Invisible team members	Club members and their roles have no public visibility on any dedicated platform.
Poor event discovery	Students miss events due to lack of an organised calendar and notification system.
No challenge ecosystem	There is no structured challenge system with real-time submission tracking and leaderboards.
Project graveyard	Club projects are rarely documented or showcased, reducing their impact and learning value.
Gallery chaos	Event photos and memories are buried in drives or group chats with no curated public archive.

3. Target Users

Club Members / Students

- Register and maintain a rich profile with skills, social links, and rank
- Participate in challenges and hackathons
- Submit solutions via GitHub and live links
- Browse the team section and connect with peers
- View personal performance analytics and badges
- Explore the event gallery and project showcase

Club Admins / Organizers

- Create and manage events (upcoming and past)
- Add challenges with difficulty, deadlines, and tags
- Manage and evaluate submissions
- Update the team section with new members and roles
- Upload gallery media and manage project entries
- Monitor real-time leaderboard and analytics

Visitors / Prospective Members

- Explore the club's identity, mission, and achievements
- Browse the team section and contact leads
- View upcoming events and how to join
- Browse project showcases and past event galleries
- Register to become a club member

4. Core Modules & Feature Specifications

The platform must implement **all** of the following modules. Each module is a separate, well-scoped unit of work evaluated independently.

4.1 Authentication & User Profiles

- Signup / Login with email and password
- OAuth integration - Google or GitHub login
- User profile: avatar, bio, social links (GitHub, LinkedIn), tech stack tags
- Role-based access control - Visitor, Member, Admin
- Personal dashboard showing joined events, challenge scores, and rank
- Account settings: password change, profile editing, notification preferences

4.2 Team Section

A dedicated, visually engaging page showcasing every member who makes the club run.

- Hierarchical display: Club Leads - Core Team - Members
- Each card: photo, name, designation, batch/year, social links
- Skills and tech-stack tags per member
- Filter by role, batch, or domain (Web, AI/ML, Design, etc.)
- Admin can add, update, or remove team members
- Animated cards with hover effects and modal detail view
- Alumni toggle to also showcase past members

4.3 Events System (Upcoming & Past)

- Unified Events page with two tabs: Upcoming Events and Finished Events
- Each event card: title, banner image, date-time, venue/mode, category tag, registration count
- Countdown timer for upcoming events
- One-click registration with confirmation
- Event detail page: full description, schedule, speakers/hosts, FAQs
- Past events show results, winner highlights, and gallery link
- Admin: create, edit, publish or unpublish events with rich text editor
- Event categories: Workshop, Hackathon, Talk, Competition, Social

4.4 Challenge & Competition System

- Challenge listing with filters: difficulty (Easy / Medium / Hard), category, status
- Each challenge: title, description, problem statement (rich text or markdown), deadline
- Submission via GitHub repo link + live hosted link
- Submission validation - URL format check, duplicate prevention
- Submission history per user with timestamps and status
- Real-time global leaderboard per challenge with score and time bonus
- Admin: create challenges, set scoring rules, review and grade submissions
- Challenge categories: UI/UX, Full-Stack, API Design, Open Innovation

4.5 Projects Showcase

A curated portfolio of club-built projects - the engine of the club's credibility.

- Project cards: title, thumbnail, description, tech stack tags, team members
- Links to GitHub repo, live demo, and project documentation
- Filter by tech stack, year, domain, or team size
- Featured or pinned projects highlighted on the homepage
- Admin: add, edit, feature, or archive projects
- Like/star system for community appreciation
- Project detail page with screenshots carousel and extended description

4.6 Media Gallery

- Masonry or grid layout with high-quality event photos
- Albums organised by event name and date
- Lightbox viewer with full-screen support
- Admin: upload images (single or bulk), assign to albums, delete
- Lazy loading for performance on low-bandwidth connections
- Optional: short video reels from events

4.7 Global Leaderboard & Performance Analytics

- Real-time global leaderboard across all challenges
- Event-wise leaderboard scoped per challenge or hackathon
- User rank badge on profile: Newbie - Contributor - Expert - Legend
- Personal analytics dashboard: score over time, participation rate, win rate
- Activity heatmap similar to a GitHub contribution graph
- Admin analytics: platform-wide stats - registrations, submissions, active users

4.8 Homepage & Club Landing

- Hero banner with club tagline and CTA buttons (Join / Explore)
- Quick stats: member count, events hosted, challenges launched, projects built
- Upcoming events preview strip
- Featured challenges and featured projects carousel
- Top leaderboard preview (Top 5)
- Club announcement ticker or notices
- Social media links and recruitment CTA

4.9 Admin Panel

- Secure admin-only route with role guard
- CRUD operations for: Events, Challenges, Team Members, Projects, Gallery Albums
- Submission review: view, score, and provide feedback on entries
- User management: promote to admin, ban account, reset password
- Platform analytics dashboard with charts for signups, registrations, and more
- Announcement or notice publisher
- Audit log of admin actions

5. Technical Requirements

Participants are free to choose any technology stack. The following requirements must be satisfied regardless of the tools or frameworks chosen:

Frontend	Responsive, mobile-first UI. Must work correctly on both desktop and mobile browsers.
Backend	RESTful or GraphQL API with secure authentication and role-based access control.
Database	Persistent data storage for users, events, challenges, submissions, and gallery assets.
File Storage	Ability to upload and serve images for gallery, profiles, and event banners.
Real-Time	Live leaderboard updates without requiring a full page refresh.
Authentication	Secure login with hashed passwords and session or token management.
Deployment	Application must be publicly accessible via a live URL with HTTPS.
Version Control	All code hosted on a public GitHub repository with a meaningful commit history.

6. Evaluation Criteria

UI / UX Design	**25%**	Visual design, responsiveness, animations, mobile-first layout
Innovation & Features	**20%**	Unique ideas, bonus features, creative problem-solving
Functionality	**20%**	All core features working, smooth user flows, no broken pages
Club-Specific Modules	**15%**	Team section, gallery, events timeline, projects showcase
Presentation & Pitching	**10%**	Clarity of explanation, confidence, live demo walkthrough, Q&A; handling
Deployment	**5%**	Live hosted application with HTTPS and fast load time
Code Quality	**5%**	Clean code, modular structure, documented README

7. Bonus Features (Optional)

AI Project / Challenge Recommendations	Suggest challenges based on user skill tags and past activity.
Badges & Achievement System	Earn badges for first submission, win streaks, and milestones.
Streak / Activity System	Daily login streak, weekly challenge streak displayed on profile.
Dark Mode	System-aware or manual dark/light toggle throughout the platform.
Email / Push Notifications	Reminders for events, challenge deadlines, and announcements.
Progressive Web App (PWA)	Installable, offline-capable mobile experience.
Alumni Network Section	Showcase past members and their current roles and companies.
Blog / Technical Articles	Club members can publish posts or tutorials on the platform.
Certificate Generation	Auto-generate participation or winner certificates as downloadable PDFs.
Multi-Language Support	Internationalisation - at least English plus one regional language.

8. Event Schedule

8th March | 9:30 AM	**Event Begins — Coding & Development Phase Starts**
8th March | Overnight	Continuous Development — Build, Test, Deploy
9th March | 12:30 PM	**Submission Deadline — GitHub link + Live URL due**
9th March | 12:30 – 2:00 PM	Lunch Break
9th March | 2:00 PM	**Shortlist Announcement — Selected teams revealed**
9th March | 2:00 – 3:30 PM	Project Presentations — Live website & code walkthrough
9th March | 3:30 PM	**Evaluation Closes — Results & Closing Ceremony**

** PPT slides and demo videos are **optional**. Judging is based on your live website and a direct code walkthrough with the team.*

9. Submission Requirements

Live Deployed Application	A publicly accessible URL (HTTPS preferred). Must be live at the time of judging by 12:30 PM on Day 2.
GitHub Repository	Public repo with a descriptive commit history. No single init dump commit. Submit link before deadline.
README Documentation	Setup instructions, feature list, tech stack, screenshots, team members, and known issues.
Demo Video (Optional)	Short walkthrough video if the team wishes to include one. Not mandatory for evaluation.
Presentation Deck (Optional)	Slide deck covering problem, solution, and architecture. Not required — live demo takes priority.

10. Rules & Guidelines

- Team size: **2 to 4 members** only. Solo submissions will not be accepted.
- All code must be written **during the event**. Pre-built templates must be disclosed in the README.
- Application **must be deployed** and live at the time of evaluation.
- GitHub repository must be **submitted before the deadline**. Late commits will be ignored.
- Plagiarism will result in **immediate disqualification** without appeal.
- Open-source libraries and UI component kits are allowed - declare them in the README.
- AI-generated code is allowed but must be **understood and explainable** by the team during judging.
- Admin panel must be protected - do **not** expose admin credentials in the public repository.